Program 1

In [6]:

```
#import pandas as pd
import numpy as np
data = pd.read_csv('lab1.csv')
data
features = np.array(data)[:,:-1]
features
target = np.array(data)[:,-1]
target

for i ,val in enumerate(target):
    if val == 'yes':
        specific_h = features[i].copy()
        break

print(specific_h)

for i,val in enumerate(features):
    if target[i] == 'yes':
        for x in range(len(specific_h)):
            if val[x] != specific_h[x]:
                specific_h[x] = '?'
print(specific_h)
```

```
['sunny' 'warm' 'normal' 'strong' 'warm' 'same']
['sunny' 'warm' '?' 'strong' '?' '?']
```

Program 2

In [2]:

```
import numpy as np
import pandas as pd
data = pd.read_csv('lab1.csv')
data
features=np.array(data)[:,:-1]
features
target=np.array(data)[:,-1]
target
specific_h=features[0].copy()
print("initialization of specific_h an general_h")
print(specific_h)
general_h=[["?" for i in range(len(specific_h))]for i in range(len(specific_h))]
print(general_h)
for i,h in enumerate(features):
    if target[i] =="yes":
        for x in range(len(specific_h)):

            if h[x]!=specific_h[x]:
                specific_h[x]='?'
                general_h[x][x]='?'
    if target[i]=="no":
        for x in range(len(specific_h)):
            if h[x]!=specific_h[x]:
                general_h[x][x]=specific_h[x]
            else:
                general_h[x][x]='?'
            print(specific_h,"\n")
            print(general_h,"\n")
indices=[i for i,val in enumerate(general_h) if val==['?','?','?','?','?','?']]
for i in indices:
    general_h.remove(['?','?','?','?','?','?'])
print("\nfinal specific_h:",specific_h,sep="\n")
print("final general_h:",general_h,sep="\n")
```

```
initialization of specific_h an general_h
['sunny' 'warm' 'normal' 'strong' 'warm' 'same']
[['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?',
'?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]
['sunny' 'warm' '?' 'strong' 'warm' 'same']

[['sunny', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?',
'?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

['sunny' 'warm' '?' 'strong' 'warm' 'same']

[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
'?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

['sunny' 'warm' '?' 'strong' 'warm' 'same']

[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
'?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

['sunny' 'warm' '?' 'strong' 'warm' 'same']

[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
'?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

['sunny' 'warm' '?' 'strong' 'warm' 'same']

[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
'?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?']]

['sunny' 'warm' '?' 'strong' 'warm' 'same']

[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?',
'?', '?', '?', '?'], ['?', '?', '?', '?', '?', '?'], ['?', '?', '?', '?', '?', 'same']]


final specific_h:
['sunny' 'warm' '?' 'strong' '?' '?']
final general_h:
[['sunny', '?', '?', '?', '?', '?'], ['?', 'warm', '?', '?', '?', '?']]
```

Program 3

In [11]:

```
import pandas as pd
from collections import Counter
import math

tennis= pd.read_csv('tennis.csv')
print("\n given play tennis dataset:\n \n",tennis)

def entropy(alist):
    c = Counter(x for x in alist)
    instances = len(alist)
    prob = [x / instances for x in c.values()]
    return sum( [-p*math.log(p,2) for p in prob] )

def information_gain(d,split,target):
    splitting=d.groupby(split)
    n=len(d.index)
    agent = splitting.agg({target : [entropy,lambda x: len(x)/n ] })[target]
    agent.columns = ['entropy','observations']
    newentropy = sum(agent['entropy']*agent['observations'])
    oldentropy=entropy(d[target])
    return oldentropy - newentropy

def id3(sub,target,a):
    count = Counter(x for x in sub[target])#class of YES / NO
    if len(count)==1:
        return next(iter(count))#next i/p dataset
    else:
        gain=[information_gain(sub,attr,target) for attr in a]
        print("\n gain=",gain)
        maximum = gain.index(max(gain))
        best = a[maximum]
        print("\nbest attribute=",best)
        tree={best:{}}
        remaining=[i for i in a if i != best]

        for val,subset in sub.groupby(best):
            subtree = id3(subset,target,remaining)
            tree[best][val] = subtree
        return tree

names = list(tennis.columns)
```

```
print("\nlist of attributes:",names)
names.remove("PlayTennis")
print("\npredicting attributes:",names)
tree=id3(tennis,'PlayTennis',names)
print("\n\n the resultant decision tree is:\n")
print(tree)
```

```
 given play tennis dataset:

      Outlook Temperature Humidity    Wind PlayTennis
0      Sunny         Hot     High    Weak         no
1      Sunny         Hot     High  Strong         no
2   Overcast         Hot     High    Weak        yes
3       Rain        Mild     High    Weak        yes
4       Rain        Cool   Normal    Weak        yes
5       Rain        Cool   Normal  Strong         no
6   Overcast        Cool   Normal  Strong        yes
7      Sunny        Mild     High    Weak         no
8      Sunny        Cool   Normal    Weak        yes
9       Rain        Mild   Normal    Weak        yes
10     Sunny        Mild   Normal  Strong        yes
11  Overcast        Mild     High  Strong        yes
12  Overcast         Hot   Normal    Weak        yes
13      Rain        Mild     High  Strong         no

list of attributes: ['Outlook', 'Temperature', 'Humidity', 'Wind', 'PlayTennis']

predicting attributes: ['Outlook', 'Temperature', 'Humidity', 'Wind']

 gain= [0.2467498197744391, 0.029222565658954647, 0.15183550136234136, 0.04812703040826927]

best attribute= Outlook

 gain= [0.01997309402197489, 0.01997309402197489, 0.9709505944546686]

best attribute= Wind

 gain= [0.5709505944546686, 0.9709505944546686, 0.01997309402197489]

best attribute= Humidity


 the resultant decision tree is:

{'Outlook': {'Overcast': 'yes', 'Rain': {'Wind': {'Strong': 'no', 'Weak': 'yes'}}, 'Sunny': {'Humidity': {'High': 'n
o', 'Normal': 'yes'}}}}
```

Program 4

In [4]:

```
import numpy as np
x=np.array(([2,9],[1,5],[3,6]),dtype=float)
y=np.array(([92],[86],[89]),dtype=float)
x=x/np.amax(x,axis=0)
y=y/100

def sigmoid(x):
    return 1/(1+np.exp(-x))
def derivatives_sigmoid(x):
    return x*(1-x)
epoch=7000
lr=0.1
inputlayer_neurons=2
hiddenlayer_neurons=3
output_neurons=1

wh=np.random.uniform(size=(inputlayer_neurons,hiddenlayer_neurons))
bh=np.random.uniform(size=(1,hiddenlayer_neurons))
wout=np.random.uniform(size=(hiddenlayer_neurons,output_neurons))
bout=np.random.uniform(size=(1,output_neurons))

for i in range(epoch):
    hinp1=np.dot(x,wh)
    hinp=hinp1+bh
    hlayer_act=sigmoid(hinp)
    outinp1=np.dot(hlayer_act,wout)
    outinp=outinp1+bout
    output=sigmoid(outinp)

    EO=y-output
    outgrad=derivatives_sigmoid(output)
    d_output=EO*outgrad
    EH=d_output.dot(wout.T)
    hiddengrad=derivatives_sigmoid(hlayer_act)
    d_hiddenlayer=EH*hiddengrad
    wout+=hlayer_act.T.dot(d_output)*lr

print("Input:\n"+str(x))
print("Actual output: \n"+str(y))
```

```
print("Predicted output:\n",output)
```

```
Input:
[[0.66666667 1.        ]
 [0.33333333 0.55555556]
 [1.         0.66666667]]
Actual output:
[[0.92]
 [0.86]
 [0.89]]
Predicted output:
 [[0.89652276]
 [0.87954289]
 [0.8931647 ]]
```

Program 5

In [10]:

```
import pandas as pd
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.model_selection import train_test_split
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score
from sklearn import metrics

data=pd.read_csv('textdata12.csv',names=['message','label'])
print('the dataset is',data)
print('the dimensions of the dataset',data.shape)
data['labelnum']=data.label.map({'pos':1,'neg':0})
x=data.message
y=data.labelnum
print(x)
print(y)
vectorizer=TfidfVectorizer()
data=vectorizer.fit_transform(x)
print("\n the feature of dataset:\n")
df=pd.DataFrame(data.toarray(),columns=vectorizer.get_feature_names_out())
df.head()
print("\n train test split")
xtrain,xtest,ytrain,ytest=train_test_split(data,y,test_size=0.3,random_state=42)
print("\n the total number of training data:",ytrain.shape)
print("\n the total number of test data:",ytest.shape)
clf=MultinomialNB().fit(xtrain,ytrain)
predict=clf.predict(xtest)
predicted=clf.predict(xtest)
print("\n accuracy of the classifier is",metrics.accuracy_score(ytest,predicted))
print("\n confusion matrix is\n",metrics.confusion_matrix(ytest,predicted))
print("\n classification report is\n",metrics.classification_report(ytest,predicted))
print("\n the value of precision is\n",metrics.precision_score(ytest,predicted))
print("\n the value of recall  is\n",metrics.recall_score(ytest,predicted))
```

```
the dataset is                                  message label
0                        i love sandwitch   pos
1                this is an amazing place   pos
2      i feel very good about these beers   pos
3                    this is my best work   pos
4                    what an awesome view   pos
5            i do not like this resturant   neg
6                i am tired of this stuff   neg
7                   i cant deal with this   neg
8                    he is my sworn enemy   neg
9                     my boss is horrible   neg
10               this is an awesome place   pos
11  i do not like the taste of this juice   neg
12                       i love to dance   pos
13      i am sick and tired of this place   neg
14                   what a great holiday   pos
15           that is bad locality to stay   neg
16         we will have good fun tomorrow   pos
17       i went to my enemy's house today   neg
the dimensions of the dataset (18, 2)
0                          i love sandwitch
1                 this is an amazing place
2        i feel very good about these beers
3                     this is my best work
4                     what an awesome view
5             i do not like this resturant
6                 i am tired of this stuff
7                    i cant deal with this
8                     he is my sworn enemy
9                      my boss is horrible
10                this is an awesome place
11    i do not like the taste of this juice
12                         i love to dance
13        i am sick and tired of this place
14                    what a great holiday
15            that is bad locality to stay
16          we will have good fun tomorrow
17         i went to my enemy's house today
Name: message, dtype: object
0     1
1     1
```

```
2     1
3     1
4     1
5     0
6     0
7     0
8     0
9     0
10    1
11    0
12    1
13    0
14    1
15    0
16    1
17    0
Name: labelnum, dtype: int64

 the feature of dataset:


 train test split

 the total number of training data: (12,)

 the total number of test data: (6,)

 accuracy of the classifier is 0.8333333333333334

 confusion matrix is
 [[3 0]
 [1 2]]

 classification report is
              precision    recall  f1-score   support

           0       0.75      1.00      0.86         3
           1       1.00      0.67      0.80         3

    accuracy                           0.83         6
   macro avg       0.88      0.83      0.83         6
weighted avg       0.88      0.83      0.83         6
```

```
the value of precision is
1.0

the value of recall  is
0.6666666666666666
```

Program 6

In [13]:

```
import pandas as pd
col=['Age','Gender','Familylist','Diet','LifeStyle','Cholesterol','heartDisease']
data=pd.read_csv('lab6.csv',names=col)
print(data)
from sklearn.preprocessing import LabelEncoder
encoder=LabelEncoder()
for i in range(len(col)):
    data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
X=data.iloc[:,0:6]
y=data.iloc[:,-1]
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(X,y,test_size=0.2)
from sklearn.naive_bayes import GaussianNB
clf=GaussianNB()
clf.fit(X_train,y_train)
y_pred=clf.predict(X_test)
from sklearn.metrics import confusion_matrix
print('confusion matrix',confusion_matrix(y_test,y_pred))
```

```
                  Age  Gender Familylist    Diet LifeStyle Cholesterol  \
0   SuperSeniorCitizen    Male        Yes  Medium  Sedetary        High
1   SuperSeniorCitizen  Female        Yes  Medium  Sedetary        High
2        SeniorCitizen    Male         No    High  Moderate  Borderline
3                 Teen    Male        Yes  Medium  Sedetary      Normal
4                Youth  Female        Yes    High   Athlete      Normal
5           MiddleAged    Male        Yes  Medium    Active        High
6                 Teen    Male        Yes    High  Moderate        High
7   SuperSeniorCitizen    Male        Yes  Medium  Sedetary        High
8                Youth  Female        Yes    High   Athlete      Normal
9        SeniorCitizen  Female         No    High   Athlete      Normal
10                Teen  Female         No  Medium  Moderate        High
11                Teen    Male        Yes  Medium  Sedetary      Normal
12          MiddleAged  Female         No    High   Athlete        High
13          MiddleAged    Male        Yes  Medium    Active        High
14               Youth  Female        Yes    High   Athlete  Borderline
15  SuperSeniorCitizen    Male        Yes    High   Athlete      Normal
16       SeniorCitizen  Female         No  Medium  Moderate  Borderline
17                Teen    Male        Yes  Medium  Sedetary      Normal

   heartDisease
0           Yes
1           Yes
2           Yes
3            No
4            no
5           Yes
6           Yes
7           Yes
8            No
9           Yes
10          Yes
11           No
12           No
13          Yes
14           No
15          Yes
16          Yes
17           No
confusion matrix [[0 1 0]
```

```
 [0 2 0]
 [1 0 0]]

C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
C:\Users\BITM\AppData\Local\Temp\ipykernel_8384\4048633161.py:8: DeprecationWarning: In a future version, `df.iloc[:,
i] = newvals` will attempt to set the values inplace instead of always setting a new array. To retain the old behavio
r, use either `df[df.columns[i]] = newvals` or, if columns are non-unique, `df.isetitem(i, newvals)`
  data.iloc[:,i]=encoder.fit_transform(data.iloc[:,i])
```

Program 7

In [15]:

```
import numpy as np
import pandas as pd
from matplotlib import pyplot as plt
from sklearn.mixture import GaussianMixture
from sklearn.cluster import KMeans
data=pd.read_csv('lab7.csv')
print('Input Data and Shape')
print(data.shape)
data.head()
f1=data['V1'].values
f2=data['V2'].values
X=np.array(list(zip(f1,f2)))
print("X",X)
print("Graph for whole dataset")
plt.scatter(f1,f2,c='black',s=15)
plt.show()
kmeans=KMeans(10,random_state=42)
labels=kmeans.fit(X).predict(X)
print("labels ",labels)
centroids=kmeans.cluster_centers_
print("Centriods   ",centroids)
plt.scatter(X[:,0],X[:,1],c=labels,s=40,cmap='viridis')
print("Graph using Kmeans Algorithm ")
plt.scatter(centroids[:,0],centroids[:,1],marker='*',s=200,c='#050505')
plt.show()
gmm=GaussianMixture(n_components=3).fit(X)
labels=gmm.predict(X)
probs=gmm.predict_proba(X)
size=10*probs.max(1)**3
print("Graph using Em algorithm")
plt.scatter(X[:,0],X[:,1],c=labels,s=size,cmap='viridis')
plt.show()
```

```
Input Data and Shape
(25, 2)
X [[5.1 3.5]
 [4.9 3. ]
 [4.7 3.2]
 [4.6 3.1]
 [5.  3.6]
 [5.4 3.9]
 [4.6 3.4]
 [5.  3.4]
 [4.4 2.9]
 [4.9 3.1]
 [5.4 3.7]
 [4.8 3.4]
 [4.8 3. ]
 [4.3 3. ]
 [5.8 4. ]
 [5.7 4.4]
 [5.4 3.9]
```

Program 8

In [16]:

```
import csv
import random
import math
import operator
def loadDataset(filename,split,trainingSet=[],testSet=[]):
    with open(filename) as csvfile:
        lines=csv.reader(csvfile)
        dataset=list(lines)
        for x in range(len(dataset)-1):
            for y in range(4):
                dataset[x][y]=float(dataset[x][y])
            if random.random()<split:
                trainingSet.append(dataset[x])
            else:
                testSet.append(dataset[x])
def euclideanDistance(instance1,instance2,length):
    distance=0
    for x in range(length):
        distance+=pow((instance1[x]-instance2[x]),2)
    return math.sqrt(distance)
def getNeighbors(trainingSet,testInstance,k):
    distances=[]
    length=len(testInstance)-1
    for x in range(len(trainingSet)):
        dist=euclideanDistance(testInstance,trainingSet[x],length)
        distances.append((trainingSet[x],dist))
    distances.sort(key=operator.itemgetter(1))
    neighbors=[]
    for x in range(k):
        neighbors.append(distances[x][0])
    return neighbors
def getResponse(neighbors):
    classVotes={}
    for x in range(len(neighbors)):
        response=neighbors[x][-1]
        if response in classVotes:
            classVotes[response]+=1
        else:
            classVotes[response]=1
    sortedVotes=sorted(classVotes.items(),key=operator.itemgetter(1),reverse=True)
    return sortedVotes[0][0]
```

```
def getAccuracy(testSet,predictions):
    correct=0
    for x in range(len(testSet)):
        if testSet[x][-1]==predictions[x]:
            correct+=1
    return (correct/float(len(testSet)))*100.0
def main():
    trainingSet=[]
    testSet=[]
    split=0.67
    loadDataset('iris.csv',split,trainingSet,testSet)
    print('\n Number of Training data:'+(repr(len(trainingSet))))
    print("Number of test data :"+repr(len(testSet)))
    predictions=[]
    k=3
    print("\n the predictions are: ")
    for x in range(len(testSet)):
        neighbors=getNeighbors(trainingSet,testSet[x],k)
        result=getResponse(neighbors)
        predictions.append(result)
        print('predicted= '+repr(result)+',actual='+repr(testSet[x][-1]))
    accuracy=getAccuracy(testSet,predictions)
    print("\n The accuracy is "+repr(accuracy)+'%')
main()
```

```
 Number of Training data:91
Number of test data :58

 the predictions are:
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-setosa',actual='Iris-setosa'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-virginica',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
```

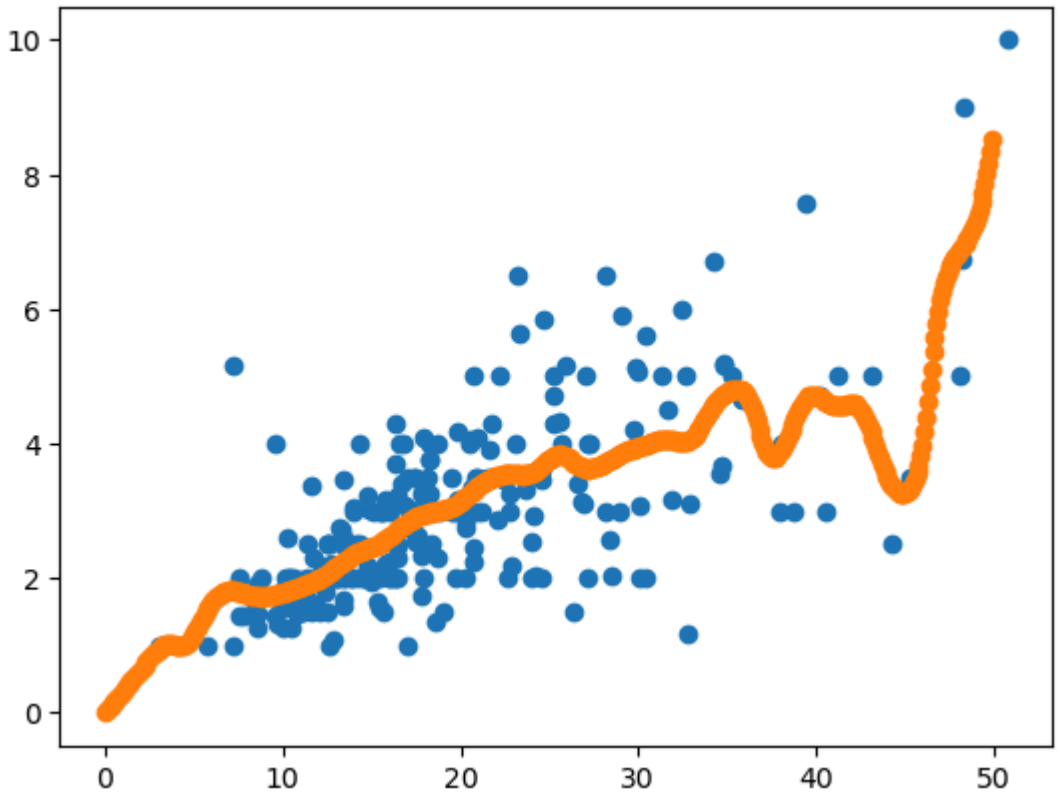
```
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-versicolor',actual='Iris-versicolor'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-versicolor',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'
predicted= 'Iris-virginica',actual='Iris-virginica'

 The accuracy is 96.55172413793103%
```

Program 9

In [17]:

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
tou=1
data=pd.read_csv("tips.csv")
X_train=np.array(data.total_bill)
print(X_train)
X_train=X_train[:,np.newaxis]
print(len(X_train))
y_train=np.array(data.tip)
X_test=np.array([i/10 for i in range(500)])
X_test=X_test[:,np.newaxis]
y_test=[]
count=0
for r in range(len(X_test)):
    wts=np.exp(-np.sum((X_train-X_test[r])**2,axis=1)/(2*tou**2))
    W=np.diag(wts)
    factor1=np.linalg.inv(X_train.T.dot(W).dot(X_train))#factor=XT.W.X
    parameters=factor1.dot(X_train.T).dot(W).dot(y_train)
    prediction=X_test[r].dot(parameters)
    y_test.append(prediction)
    count+=1
print(len(y_test))
y_test=np.array(y_test)
plt.plot(X_train.squeeze(),y_train,'o')
plt.plot(X_test.squeeze(),y_test,'o')
plt.show()
```

```
[16.99 10.34 21.01 23.68 24.59 25.29  8.77 26.88 15.04 14.78 10.27 35.26
 15.42 18.43 14.83 21.58 10.33 16.29 16.97 20.65 17.92 20.29 15.77 39.42
 19.82 17.81 13.37 12.69 21.7  19.65  9.55 18.35 15.06 20.69 17.78 24.06
 16.31 16.93 18.69 31.27 16.04 17.46 13.94  9.68 30.4  18.29 22.23 32.4
 28.55 18.04 12.54 10.29 34.81  9.94 25.56 19.49 38.01 26.41 11.24 48.27
 20.29 13.81 11.02 18.29 17.59 20.08 16.45  3.07 20.23 15.01 12.02 17.07
 26.86 25.28 14.73 10.51 17.92 27.2  22.76 17.29 19.44 16.66 10.07 32.68
 15.98 34.83 13.03 18.28 24.71 21.16 28.97 22.49  5.75 16.32 22.75 40.17
 27.28 12.03 21.01 12.46 11.35 15.38 44.3  22.42 20.92 15.36 20.49 25.21
 18.24 14.31 14.    7.25 38.07 23.95 25.71 17.31 29.93 10.65 12.43 24.08
 11.69 13.42 14.26 15.95 12.48 29.8   8.52 14.52 11.38 22.82 19.08 20.27
 11.17 12.26 18.26  8.51 10.33 14.15 16.   13.16 17.47 34.3  41.19 27.05
 16.43  8.35 18.64 11.87  9.78  7.51 14.07 13.13 17.26 24.55 19.77 29.85
 48.17 25.   13.39 16.49 21.5  12.66 16.21 13.81 17.51 24.52 20.76 31.71
 10.59 10.63 50.81 15.81  7.25 31.85 16.82 32.9  17.89 14.48  9.6  34.63
 34.65 23.33 45.35 23.17 40.55 20.69 20.9  30.46 18.15 23.1  15.69 19.81
 28.44 15.48 16.58  7.56 10.34 43.11 13.   13.51 18.71 12.74 13.   16.4
 20.53 16.47 26.59 38.73 24.27 12.76 30.06 25.89 48.33 13.27 28.17 12.9
 28.15 11.59  7.74 30.14 12.16 13.42  8.58 15.98 13.42 16.27 10.09 20.45
 13.28 22.12 24.01 15.69 11.61 10.77 15.53 10.07 12.6  32.83 35.83 29.03
 27.18 22.67 17.82 18.78]
244
500
```

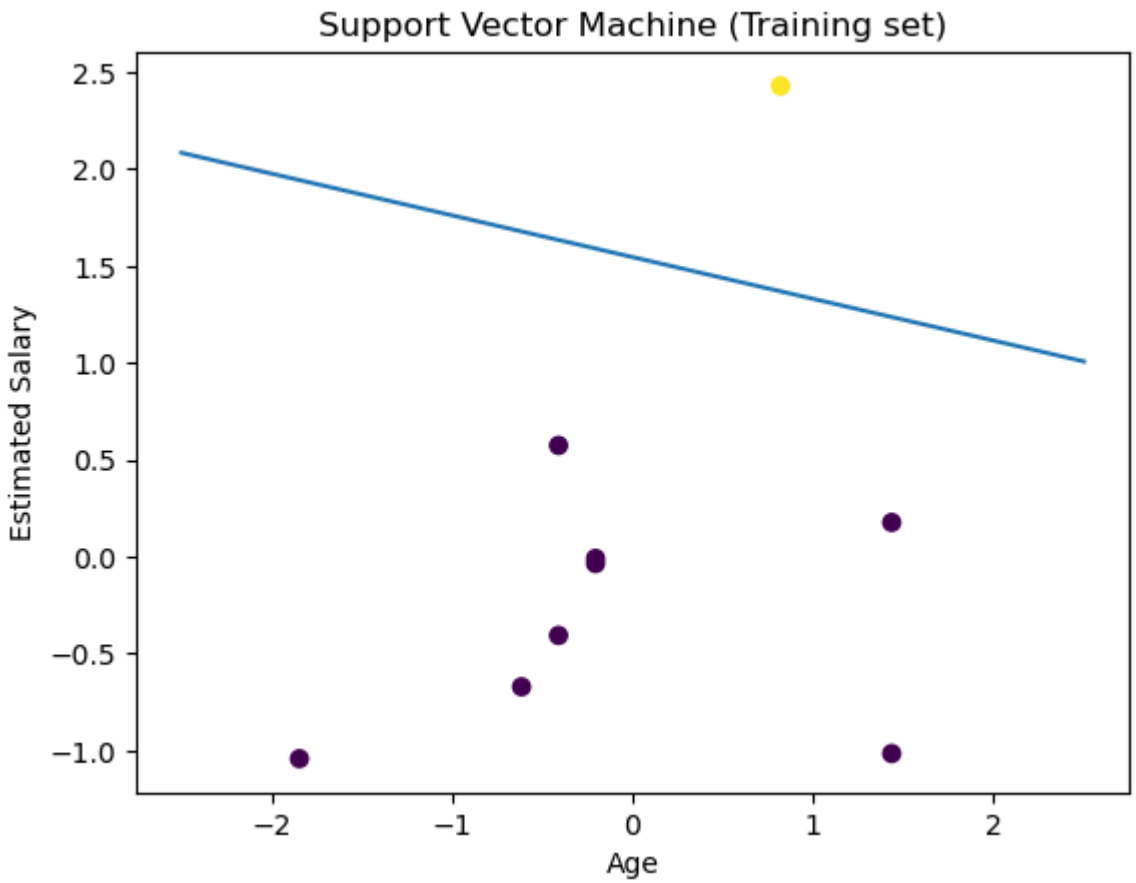

10
8
6
4
2
0
0
10
20
30
40
50

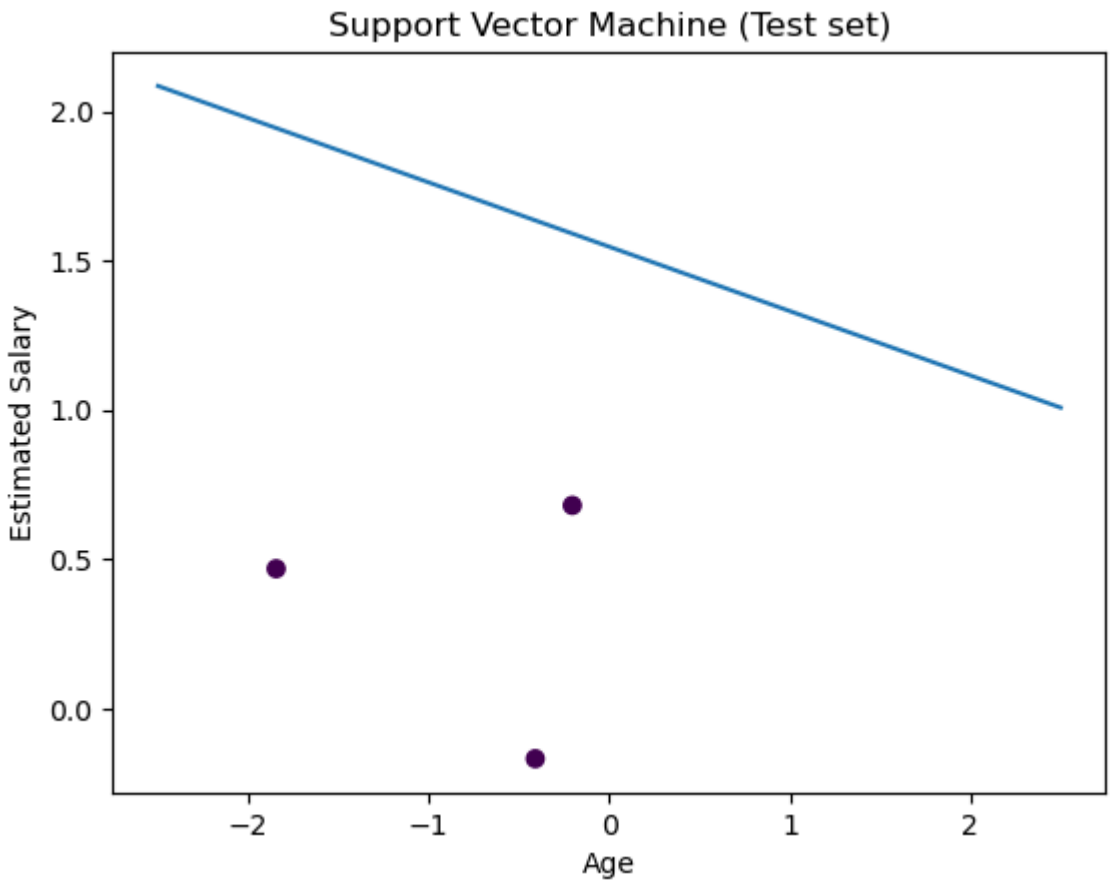
In [1]:

```
import numpy as np
import matplotlib.pyplot as plt
import pandas as pd
datasets = pd.read_csv('lab10.csv')
X = datasets.iloc[:, [2,3]].values
Y = datasets.iloc[:, 4].values
from sklearn.model_selection import train_test_split
X_Train, X_Test, Y_Train, Y_Test = train_test_split(X, Y, test_size = 0.25,
random_state = 0)
from sklearn.preprocessing import StandardScaler
sc_X = StandardScaler()
X_Train = sc_X.fit_transform(X_Train)
X_Test = sc_X.transform(X_Test)
from sklearn.svm import SVC
classifier = SVC(kernel = 'linear', random_state = 0)
classifier.fit(X_Train, Y_Train)
Y_Pred = classifier.predict(X_Test)
from sklearn import metrics
print("Accuracy score ",metrics.accuracy_score(Y_Test, Y_Pred))
plt.scatter(X_Train[:,0], X_Train[:, 1],c=Y_Train)
plt.title('Support Vector Machine (Training set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
w=classifier.coef_[0]
a=-w[0]/w[1]
xx=np.linspace(-2.5,2.5)
yy=a*xx -(classifier.intercept_[0])/w[1]
plt.plot(xx,yy)
plt.show();
plt.scatter(X_Test[:,0], X_Test[:, 1],c=Y_Test)
plt.title('Support Vector Machine (Test set)')
plt.xlabel('Age')
plt.ylabel('Estimated Salary')
w=classifier.coef_[0]
a=-w[0]/w[1]
xx=np.linspace(-2.5,2.5)
yy=a*xx -(classifier.intercept_[0])/w[1]
plt.plot(xx,yy)
plt.show()
```

```
40
```

```
Accuracy score  1.0
```




Support Vector Machine (Test set)
Estimated Salary
Age
2.0
1.5
1.0
0.5
0.0
−2
−1
0
1
2

In [ ]:

```
1
```